

Task - 8: SQL Injection Practical Exploitation

Internship Task – Elevate Labs

This task focused on understanding SQL Injection attacks and performing controlled exploitation using automated tools. The goal was to learn how database vulnerabilities are identified, exploited, and secured in web applications.

1. Introduction to SQL Injection

SQL Injection is a web application vulnerability where attackers manipulate SQL queries by injecting malicious input. This can lead to unauthorized access to databases.

2. Identifying Injectable Parameters

User input fields and URL parameters were analyzed to identify potential injection points. Improper input validation increases the risk of SQL Injection.

3. Using SQLMap for Exploitation

SQLMap was used to automate the detection and exploitation process. It identifies vulnerable parameters and confirms SQL Injection vulnerabilities.

4. Extracting Database Information

Using SQLMap, database names were enumerated, followed by extraction of tables and available data. This demonstrates the severity of database compromise.

5. User Data Extraction

Sensitive information such as usernames and passwords can be accessed if SQL Injection vulnerabilities exist.

6. Impact Analysis

Successful SQL Injection attacks can result in: - Data breaches - Unauthorized access - Data modification or deletion

7. Attack Flow Documentation

The attack process was documented step-by-step, from identifying parameters to extracting database information.

8. Mitigation and Fixes

- Use prepared statements - Validate and sanitize user input - Apply least privilege to database users - Implement Web Application Firewalls

Learning Outcome

This task improved my understanding of SQL Injection attacks and database security. I learned how attackers exploit vulnerabilities and how developers can prevent such attacks using secure coding practices.